

Modulation of the Leading-Edge Vortex Shedding Rate in Discrete-Vortex Methods

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A new method to determine the rate of circulation fed into a leading-edge vortex (LEV) is presented for discrete-vortex methods (DVM). The strength of vortex particles shed from the leading edge is calculated based on the velocity at the shear layer, which is formulated in terms of an inviscid parameter from unsteady thin-airfoil theory (UTAT), the leading-edge suction parameter (LESP). This new model is compared against a widely established method to determine the best way of simulating leading-edge vortex shedding in low-order models. Computational fluid dynamics (CFD) simulations and experiments are combined to test the performance of both models in two distinctive scenarios dominated by large-scale vortices: rapid transient pitch up maneuvers at a Reynolds number as low as $Re=10000$ and classic dynamic stall studies, characterized by periodic oscillations around the static stall angle at a high Reynolds number of 550000 . The improved performance of the new model is particularly evident when the effects of thickness and Reynolds number are important.

Nomenclature

A_0	=	zero-th Fourier coefficient
A_n	=	n-th Fourier coefficient
C_D	=	drag coefficient
C_L	=	lift coefficient
C_M	=	moment coefficient
C_S	=	suction force coefficient
c	=	airfoil chord length
c_{pvt}	=	pivot axis position
K_α	=	reduced pitch rate (ramp), $K_\alpha = \frac{\dot{\alpha}c}{2U}$
k	=	reduced frequency (harmonic), $k = \frac{\omega c}{2U}$
Re	=	chord-based Reynolds number, $\frac{Uc}{\nu}$
r_{LE}	=	airfoil leading-edge radius
T	=	period
t	=	time
t^*	=	convective time, $t^* = \frac{tU}{c}$
U	=	free stream velocity
u_{LE}	=	leading-edge velocity
W	=	local downwash

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α	=	pitch angle
$\dot{\alpha}$	=	pitch rate
Γ_B	=	bound circulation
Γ_{LEV}	=	leading-edge vortex total circulation
Γ_{TEV}	=	trailing-edge vortex total circulation
Γ_i	=	strength of the i-th leading-edge discrete vortex
$\dot{\Gamma}_{LE}$	=	growth rate of circulation at the leading edge
γ	=	bound vorticity distribution
η	=	normal coordinate
η_c	=	airfoil camber
θ	=	transformation variable
ξ	=	chordwise coordinate
ϕ_{LEV}	=	velocity potential from leading-edge vorticity
ϕ_{TEV}	=	velocity potential from trailing-edge vorticity
ω	=	angular frequency

I. Introduction

Dynamic events in fluids, such as the formation and development of vortices around aerodynamic surfaces, are frequently visible in nature and of scientific interest to a large number of disciplines [1]. Fluid dynamical analysis of these flows typically focuses on the evolution of vorticity fields. In this sense, the continuous distribution of vorticity in a shear layer can be replaced with a finite array of computational elements, each representing a discrete concentration of vorticity. By tracking their translation and deformation a full description of the flow field is obtained. This is the basis of vortex methods [2, 3]. The close link of these numerical methods with the physics they aim to reproduce, and their adaptability to changing conditions in the flow, justify why they are exploited to simulate separated flows. Several endeavors to numerically compute the unsteady motion of two-dimensional vortex sheets followed Rosenhead's pioneering research in the 1930's [4]. Among these examples are the roll-up of free vortex sheets [5], or the separation and vortex shedding in flows past edges at steady motion, like bluff bodies [6], inclined flat plates [7], or cambered airfoils where the separation point is behind the leading edge [8]. An excellent historical background on these methods can be found in [9, 10], and a review of improvements in accuracy and computational efficiency is provided by [11].

Further insight into the physics of flow separation was gained by investigations from the rotorcraft community through a phenomenon known as dynamic stall. Aerodynamic bodies executing unsteady motions can delay the onset of stall to incidences far beyond their static stall limits [12]. The distinguishing feature of this process is the formation of a vortex at the leading edge, which traverses the chord and ultimately sheds [13]. The phenomenon was first identified on helicopter rotor blades, and has been a major concern since. Albeit the airfoil experiences a lift enhancement while the dynamic-stall vortex (DSV) stays attached, the benefit does not come without risk, as fatigue and structural damage can be a consequence of undesirable load fluctuations that are associated with the shedding of the stall vortex. Being an important limiting factor in the operation of rotary wings, engineers in this field aspire to prevent dynamic stall, and many semi-empirical methods to model the phenomenon have been developed as a result [14]. Most of the research on this subject considers oscillatory pitch kinematics at medium to high Reynolds numbers and low frequencies, consistent with the movement of rotor blades.

In a quest to understand the enigmatic nature of flapping-wing flight, biologists revealed that it relied on unsteady flow separation and a strong leading-edge vortex [15–18]. The acknowledged benefit that these vortices confer to natural fliers has sparked a growing interest within the aerospace community and motivated them to develop highly-maneuverable insect-size flying devices or micro aerial vehicles (MAV) [19]. In contrast to rotary-wing mechanisms, flapping fliers operate at high dimensionless rates of motion and low Reynolds numbers. Although the chronology of events associated with leading-edge separation and vortex formation is generally the same for most unsteady airfoils, review of the literature reveals that the characteristics of the vortical structures depend on many parameters - reduced frequency, Reynolds number [20, 21] - and the type of motion involved. To better comprehend the LEV behavior on airfoils operating at these flight regimes, either to stabilize the vortex and benefit from its temporary lift boost, or to alleviate transients in gust encounters, mathematical models are sought after that capture the essential physics of the unsteady flow problem in a simple way [18, 22].

With an increase in computing power and a wider access to resources, the use of vortex methods has received a new impetus in the last decade. Low-order models employing discrete vortices are being developed for fast simulations of

unsteady flows with application to bio-inspired flight [23–27]. Built upon potential-flow theories [28–30], these models might be augmented to account for non-linearities due to viscous effects, which are prevalent in these flight regimes. A review of early investigations aiming to extend the scope of classical inviscid-fluid theories to account for viscous phenomena, like separation and stall, is given in [31]. To be successful in modeling unsteady flow problems, low-order methods need to incorporate augmented theories. Some recent efforts in this direction have focused on determining the appropriate release of vorticity into the flow field through free vortex sheets formed at both airfoil edges. At the trailing edge, this has been achieved by extensions to the classical steady Kutta condition [32–34]. Because there is evidence of a correlation between critical events at the leading edge and the initiation of local reverse flow [35–38], criteria to predict the onset of unsteady separation have been developed based on different leading-edge flow quantities: leading-edge suction [39, 40]; boundary-layer characteristics [41]; or wall shear stress [42].

Among these stall onset criteria, the one based on the critical leading-edge suction parameter, which served Ramesh et al. to develop the LESP-modulated discrete-vortex method (LDVM) [39], has been extensively adopted in the development of discrete-vortex models [43–48]. The idea behind this criterion is that a rounded leading edge can support a certain limit of suction, for a given Reynolds number, over which the flow detaches. Even during the vortex-shedding process, suction is limited to that maximum amount. However, recent investigations have shown that suction force at the leading edge decays during the formation of an LEV [49–51]. It is the aim of this paper to improve the post-stall behavior of low-order models augmented to capture LEV formation. For this purpose, a new approach in the calculation of the strength of discrete vortices is introduced. The new formulation is based on the velocity at the separated shear layer, which is expressed in terms of a parameter from unsteady-thin-airfoil theory.

The structure of the paper is as follows: Sec. II elaborates on the mathematical foundations of the model; in Sec. III details on the CFD configuration are provided; the experimental set-up is explained in Sec. IV; results for three different cases are presented in Sec. V, including extensive comments on advantages and limitations of the model; finally, conclusions are drawn in Sec. VI.

II. Theoretical model

This first section introduces the basics of the mathematical model used in this work: the potential-flow theory adapted for unsteady airfoils on which it is built; the time-stepping scheme implemented in the discrete-vortex method; the criterion adopted to determine the onset of leading-edge vortex formation; and the different approaches considered to calculate the vorticity shedding rate from the separation point.

A. Unsteady-Thin-Airfoil Theory

We assume an inviscid and incompressible flow. The hypothesis stating inviscid flow may seem inappropriate for studies of the aerodynamics at low Reynolds number, since strong viscous effects are dominant in these regimes. However, the representation of physical phenomena like leading-edge vortices is still possible within the framework of these assumptions. This is accomplished by empirically augmenting the potential-flow theory.

In unsteady thin-airfoil theory, the solution is obtained by distributing singularities over the surface on which a zero-normal-flow boundary condition is enforced. Point vortices are chosen as elementary solutions. The expression of the bound vorticity distribution is written in transformed coordinates as a Fourier series, which implicitly enforces the zero vorticity Kutta condition at the trailing edge,

$$\gamma(\theta, t) = 2U \left[A_0(t) \frac{1 + \cos \theta}{\sin \theta} + \sum_{n=1}^{\infty} A_n(t) \sin(n\theta) \right] \quad (1)$$

where U is the velocity of the free stream; $A_0, \dots, A_n$ the time-dependent coefficients of the Fourier series; and θ the transformation variable expressed in a body frame of reference by

$$\xi = \frac{c}{2}(1 - \cos \theta) \quad (2)$$

here ξ represents the chordwise coordinate, and c is the chord length.

Fourier coefficients are defined as functions of the instantaneous local downwash on the airfoil, W , according to

$$A_0(t) = -\frac{1}{\pi} \int_0^\pi \frac{W(\xi, t)}{U} d\theta \quad (3a)$$

$$A_n(t) = \frac{2}{\pi} \int_0^\pi \frac{W(\xi, t)}{U} \cos(n\theta) d\theta \quad . \quad (3b)$$

Assuming small-disturbance in the flow, the boundary condition that it must remain tangential to the surface is transferred from the camberline to the chordline. The instantaneous local downwash, which contains the contributions of motion kinematics and vortices-induced velocities, is given for a general flapping airfoil by

$$\begin{aligned} W(\xi, t) = & \frac{\partial \eta_c}{\partial \xi} \left(U \cos \alpha + \dot{h} \sin \alpha + \frac{\partial \phi_{TEV}}{\partial \xi} + \frac{\partial \phi_{LEV}}{\partial \xi} \right) \\ & - U \sin \alpha + \dot{h} \cos \alpha - \dot{\alpha} (\xi - c_{pvt}) - \frac{\partial \phi_{TEV}}{\partial \eta} - \frac{\partial \phi_{LEV}}{\partial \eta} \end{aligned} \quad (4)$$

where η represents the normal coordinate in the body frame of reference; $\eta_c(\xi)$ is the camber distribution on the airfoil; kinematics parameters are the plunge velocity, $\dot{h}$, the pitch angle, α , and the pitch rate, $\dot{\alpha}$; the position of the pivot axis is indicated by c_{pvt} ; and $\frac{\partial \phi_{TEV}}{\partial \xi}$, $\frac{\partial \phi_{TEV}}{\partial \eta}$, $\frac{\partial \phi_{LEV}}{\partial \xi}$, $\frac{\partial \phi_{LEV}}{\partial \eta}$ are the induced velocities by trailing- and leading-edge discrete vortices tangential and normal to the chord.

The bound circulation can be obtained by integrating the distribution of vorticity along the airfoil chord,

$$\Gamma_B(t) = U c \pi \left(A_0(t) + \frac{A_1(t)}{2} \right) \quad . \quad (5)$$

B. Discrete-vortex time-stepping scheme

The time-stepping approach described in [52] is followed in the construction of the present model. According to this, a discrete vortex is shed from the trailing edge at each time step, as the airfoil moves along a prescribed trajectory. The strength of shed discrete vortices is computed by an iterative process from the relationship between the bound vortex sheet and the free vortex sheets formed at both airfoil edges, expressed through the Kelvin circulation theorem:

$$\frac{d\Gamma_B(t)}{dt} + \frac{d\Gamma_{LEV}(t)}{dt} + \frac{d\Gamma_{TEV}(t)}{dt} = 0 \quad . \quad (6)$$

Shed particles are then convected downstream with the free stream velocity, and roll-up into vortical structures is possible through mutual interactions. Details of the scheme can be found in [23, 39].

C. Leading-edge vortex formation criterion

Central to the study of vortex shedding from the leading edge is the zero-th term in the Fourier expansion of the bound vorticity distribution, A_0 .

Inspired by the knowledge that leading-edge separation is strongly connected to critical flow events in this region [35–38], Ramesh et al. [39] sought for an inviscid parameter, which could be determined at every time step of a numerical method, that would allow to predict viscous behavior like LEV formation. The leading-edge suction parameter, which is a non-dimensional measure of the suction at the leading edge, was introduced for this goal. It was hypothesized that the LESP would be related to the A_0 term:

$$LESP(t) = A_0(t) \quad . \quad (7)$$

Furthermore, it was suggested that a critical value of LESP would indicate the onset of LEV formation. This value is kinematics-independent for a given airfoil and Reynolds number, as long as the motion does not result in significant trailing-edge flow separation [53].

D. Leading-edge vortex shedding rate

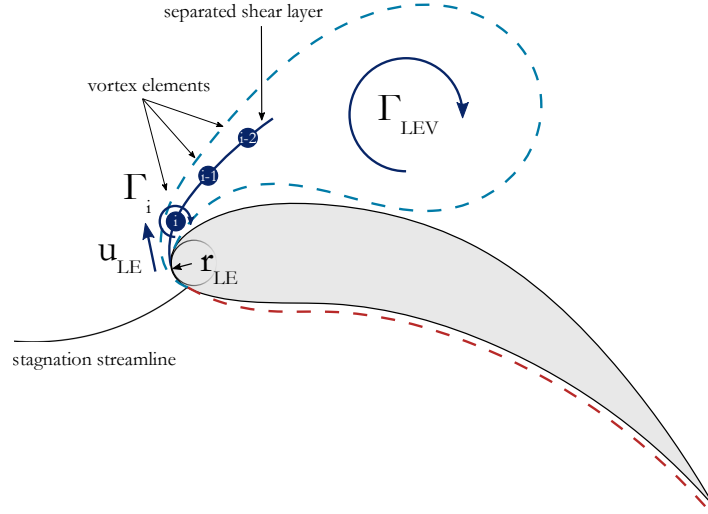


Fig. 1 Schematic illustrating a separated shear layer at the leading edge represented by vortex elements, along with the parameters involved in the strength calculation.

The success of vortex methods in modeling the evolution of vortical structures relies on the accuracy with which the rate of vorticity transport through the feeding shear layer is captured. Since the separated vortex sheet is treated as an array of vortex elements, the strength of nascent vortices on the leading edge must be properly determined at each time interval. We elaborate next on two different approaches to tackle the problem.

1. Constant A_0 -model

In previous work Ramesh et al. [39] postulated that the critical value of the LESP remains fixed during LEV shedding. When the instantaneous theoretical A_0 would exceed this critical value, due to the motion or vortex-induced velocities, vorticity is shed at the leading edge in the form of a discrete vortex. The strength of this discrete vortex is determined by requiring that the instantaneous LESP equal the critical value. The shedding process terminates when the instantaneous A_0 falls below that limit. For the cases analyzed in this paper: $LESP_{crit} = 0.12$ for Sec. V.A, and $LESP_{crit} = 0.29$ for Sec. V.B.

This original implementation of LDVM will be referred to as "*constant A_0* " approach hereafter.

2. Variable A_0 -model

In this paper, we present an alternative method to determine the rate of vorticity released from the separation point, here the leading edge. We recover an expression suggested in the literature [6–8, 54] to estimate the vorticity flux through a separated shear layer in terms of the velocity at its outside edges:

$$\frac{d\Gamma_s}{dt} = \frac{1}{2}(V_1^2 - V_2^2) \quad (8)$$

where V_1 and V_2 are the velocities at both edges of the shear layer.

This strategy to determine the rate of vorticity shed from a separation point was named the shear velocity method in the literature [8]. If the separation point over the airfoil's surface is placed at the leading edge, as it is generally assumed in unsteady flow problems where the LEV is the dominant feature, Eq. 8 can be simplified to

$$\dot{\Gamma}_{LE} = \frac{1}{2}u_{LE}^2 \quad (9)$$

Now, one of the main challenges in implementing this expression on simplified models based on thin-airfoil theory is the inherent singularity of primitive variables at the leading edge. The A_0 term of the bound vorticity distribution in

UTAT, which represents the suction peak, is infinite at the leading edge. The velocity is also infinite at this location.

A resolution to the problem is due to Ramesh [55], who applied the principle of Matched Asymptotic Expansions (MAE) to derive an expression for the velocity at the leading edge dependent on only A_0 . To resolve the singularity, he matched the solution from UTAT (outer solution) with the solution from unsteady potential flow past a parabola (inner solution), to obtain

$$u_{LE} = \sqrt{\frac{2}{r_{LE}}} U A_0 \quad (10)$$

where r_{LE} is the radius of the airfoil at the leading edge. For a symmetrical NACA airfoil, it is defined as $r_{LE} = 1.1019t^2$, with t here being the thickness.

By substituting Eq. 10 in Eq. 9 we come to a final expression for the growth rate of circulation at the leading edge, which results to be proportional to the square of the A_0 coefficient, and inversely proportional to the leading-edge radius,

$$\dot{\Gamma}_{LE} = \frac{U^2 A_0^2}{r_{LE}} \quad (11)$$

Calculating the strength of new discrete vortices entering the flow field from the leading edge, as illustrated in Fig. 1, is therefore immediate:

$$\Gamma_i = \frac{U^2 A_0^2}{r_{LE}} \Delta t \quad (12)$$

The implementation of LDVM with this scheme, which does not restrict the instantaneous LESP value during the shedding process, will be labeled "*variable A_0* " in subsequent sections.

III. Computational Fluid Dynamics model

An in-house version of the computational fluid dynamics open-source software, OpenFOAM, is used to validate the low-order model results in Sec. V.A and Sec. V.B. A body-fitted mesh was applied to the airfoil, allowing motion according to prescribed kinematics. A finite volume method (FVM) was used to solve the time-dependent incompressible Navier-Stokes equations. A second-order backward implicit scheme is used to discretize the time derivatives, and second-order limited Gaussian integration schemes are used for the gradient, divergence, and Laplacian terms. Pressure-velocity coupling is achieved using the pressure implicit with splitting of operators (PISO) algorithm. For turbulence closure the Spalart-Allmaras turbulence model [56] was chosen because of its extensive and successful use in unsteady separated flow problems. Unsteady flow simulations were performed at a low Reynolds number of $Re = 10^4$, using a chord length of 3 inches and a freestream velocity of 0.1312 m/s.

O-meshes are constructed with 246 cells around the airfoil (with a fine resolution near the leading and trailing edges), 257 cells in the wall-normal direction, and with the farfield extending to 12 chord lengths in all directions from the airfoil. No-slip boundary condition is enforced on the airfoil surface, and for the far field, freestream (inlet/outlet) boundary conditions are used. This behaves as a zero-gradient condition when fluid is flowing out of the boundary face, and as a fixed value condition (equal to freestream) when fluid is not flowing out.

IV. Experimental setup

Our model will be further validated using experimental data for a sinusoidal pitching NACA0015 airfoil [51]. The airfoil was mounted in the closed-circuit, low-speed wind tunnel of the German aerospace center in Göttingen. The incoming free stream velocity was at $U_\infty = 30 \text{ m s}^{-1}$. The chord length was $c = 0.3 \text{ m}$, leading to a Reynolds number based on the chord of $Re = 5.5 \times 10^5$. The airfoil was pitched around the quarter chord axis by a servomotor. The airfoil was pitched around the static stall angle of attack $\alpha_0 = 20^\circ$ with an amplitude of $\alpha_1 = 8^\circ$ and frequency of f_{osc} . The reduced frequency k for the airfoil ranges from 0.025 to 0.15.

The airfoil surface pressure distribution was measured using 36 pressure transducers integrated in the model mid-span cross section. Lift and pitching moment histories were obtained by integration of the pressure distribution around the airfoil. The sampling rate for the pressure measurement was 20 kHz. Two-dimensional time-resolved particle image velocimetry (TR-PIV) was used to measure the flow field in the vertical plane at the model mid-span. The flow field was illuminated by a dual-pulsed laser and the particle images were recorded by two PCO DIMAX HS cameras

with $1800 \text{ px} \times 1600 \text{ px}$ sensors. The first camera was zoomed in near the leading edge and the second camera had a larger field of view covering the rest of the airfoil. The sampling rate for the PIV was 1.5 kHz. A multi-grid algorithm was used to obtain the velocity vectors with a final window size of $48 \text{ px} \times 48 \text{ px}$ and 50 % overlap.

V. Results

The amount of leading-edge suction an airfoil can support depends on its geometry and the Reynolds number at which it operates. To examine the dependency on these two parameters, and to systematically increase the complexity of the problem, the following cases have been defined:

- i) A thin airfoil performing a transient maneuver at high rate of motion and low Reynolds number, to validate the new model against the original LDVM;
- ii) A thick airfoil undergoing the same kinematics at the same Reynolds number, to prove the expected improvement of the new model in including the effect of thickness for the calculation of the leading-edge vorticity shedding rate;
- iii) An even thicker airfoil oscillating harmonically at low reduced frequencies and high Reynolds number, a classic scenario representative of dynamic stall on rotary blades. This last case will serve to explore the behavior in a more challenging scenario as most low-order models are built with the hypothesis of no trailing-edge separation.

Details of the parameters defining each motion are included in the respective subsections.

A. Thin airfoil at low Reynolds number: NACA0004, $\text{Re} = 10000$

The first case is of interest to biological flight and the design of micro-aerial vehicles, where thin wings have to perform rapid maneuvers in viscous-dominated regimes. A symmetric NACA0004 airfoil pitches around its quarter-chord according to a smoothed ramp-hold-return motion defined by [57]:

$$\alpha(t) = \frac{K_\alpha}{a} \ln \left[\frac{\cosh(a(t - t_1)) \cosh(a(t - t_3))}{\cosh(a(t - t_2)) \cosh(a(t - t_4))} \right] \quad (13)$$

where K_α is the reduced pitch rate; the parameter a determines the smoothing at the transition points of the motion; t_1 is the time from a reference until start of ramp, $t_2 = t_1 + \alpha_{\max}/\dot{\alpha}$, $t_3 = t_2 + t_{\text{hold}}$, $t_4 = t_3 + \alpha_{\max}/\dot{\alpha}$.

For the case presented $K_\alpha = 0.2$ and $\alpha_{\max} = 45^\circ$.

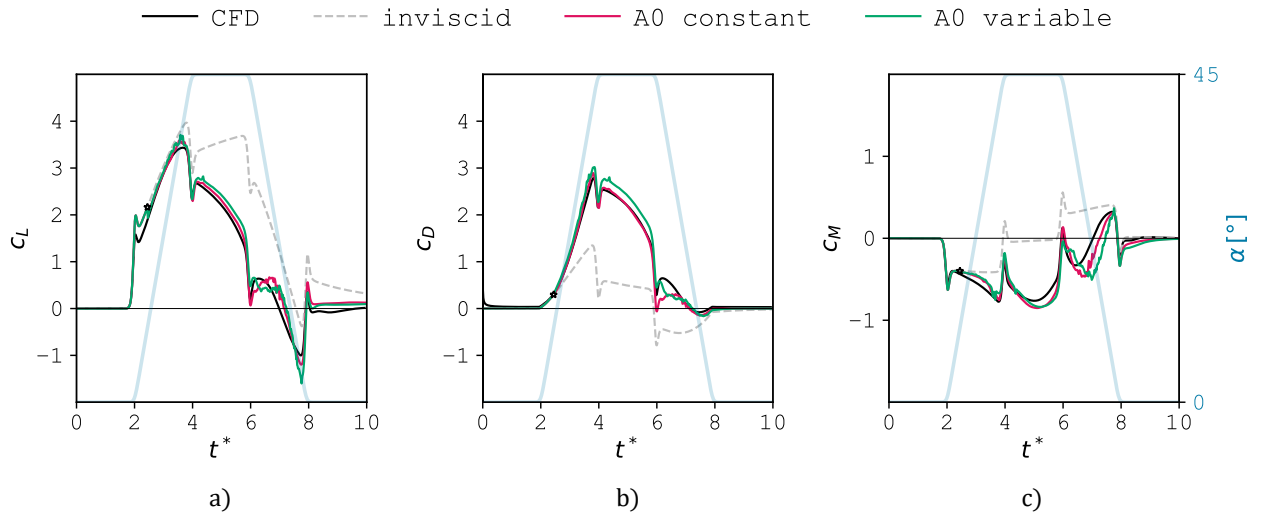


Fig. 2 Case I: aerodynamic coefficients. a) lift coefficient, b) drag coefficient, c) quarter-chord moment coefficient. A black marker indicates the instant of LEV formation. The pitch angle profile is also plotted for reference.

The aerodynamic coefficients predicted by LDVM, and the results obtained from CFD simulations are presented in

Fig. 2 versus the convective time $t^* = tU/c$. The two approaches to determine the strength of vortex elements differ little in forces and moment time histories when compared to the high-fidelity simulations. Added-mass effects, in the form of clear spikes when the angular acceleration is large, are correctly captured by both models. The magnitude of all coefficients is in good agreement during the entire time series, aside from the moment coefficient in the downstroke part of the motion, where LDVM predicts a slightly more negative value than the CFD calculation. Overall, both approaches model the feeding of vorticity into the LEV well and effectively estimate aerodynamic loads for thin airfoils at low Reynolds number.

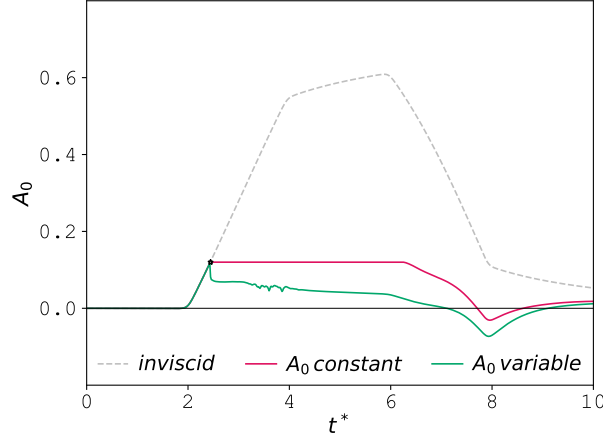


Fig. 3 Case I: Leading-edge suction parameter time history.

The original criterion for LEV-modeling assumes a constant LESP critical value during the formation of the vortex [39]. This is a valid assumption in many cases. But recently, through computation [49] and experiments [50, 51], it was revealed that the suction force during the formation of a leading-edge vortex does not hold fixed at the determined critical value, as hypothesized in the original version of LDVM, but experiences a sharp decline instead.

But how does the instantaneous value of the leading-edge suction parameter vary during LEV shedding is still a matter of study [40, 50, 51, 58]. It is important to make a distinction between the different definitions of leading-edge suction parameter found in the literature. While Ramesh et al. [39] relates LESP to the instantaneous value of the A_0 coefficient from unsteady-thin-airfoil-theory; and Narsipur et al. [40] offers an updated version of LESP incorporating a velocity correction for non-dimensionalization of A_0 ; in experimental research [50, 51, 58] this parameter is defined as the chordwise projection of the suction force vector. Therefore, in spite of being them all strongly related to A_0 , a direct comparison is not straightforward.

For the two schemes presented in this work, Fig. 3 illustrates the post-separation behavior of the leading-edge suction parameter as defined in LDVM, this is the zero-th Fourier term from UTAT. Prior to flow separation, both schemes exhibit a similar behavior. At the onset of LEV formation, marked with a black star, a sudden drop in LESP occurs in the *variable* A_0 approach; immediately after, the instantaneous value progressively departs from the *constant* A_0 curve, since the leading-edge suction decreases during the LEV formation stage [50]. This is captured with the new approach. Once the airfoil pitches back down and the flow begins to reattach at the leading edge, the deviation between models becomes smaller, until they finally collapse by the end of the motion.

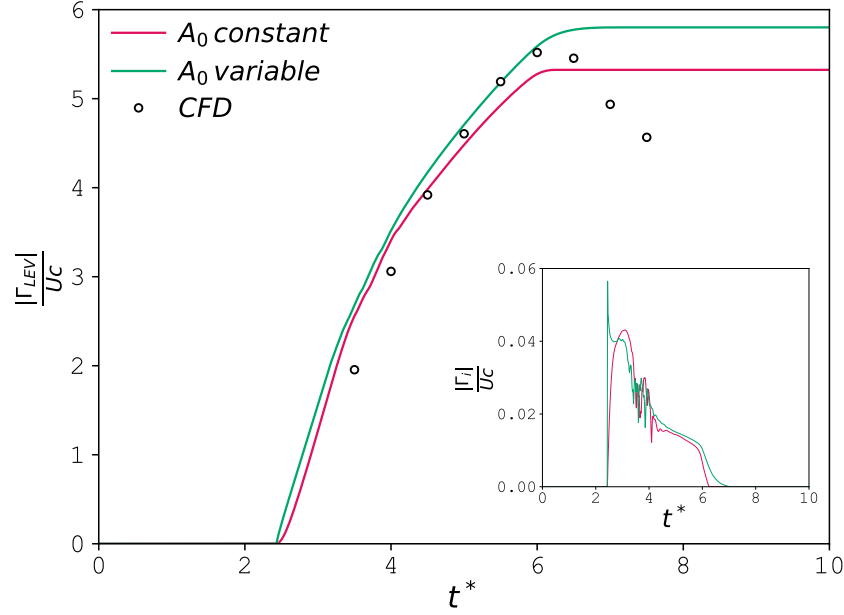


Fig. 4 Case I: Leading-edge vortex total circulation. Inset: nascent leading-edge discrete vortex strength.

The separated shear layer is the feeding mechanism of the LEV. The strength assigned to discrete vortices representing this sheet is directly connected to the vortex growth stage. This relation can be quantified by looking at the LEV circulation, which is displayed in Fig. 4. The inset graph provides information of the calculated discrete-vortex strength at each time step during LEV shedding. The way to determine the strength of shed vortices is not the same for the *constant* A_0 approach and for the *variable* A_0 approach. For the *constant* A_0 approach, the strength of shed vortices is proportional to the difference between the instantaneous inviscid A_0 and the critical value. For the *variable* A_0 approach, the strength is proportional to the instantaneous value of A_0 squared. For the specific case analyzed in this section the magnitude is nearly equal for both LDVM implementations. First, a maximum is reached soon after the onset of leading-edge vortex formation. It follows a sharp decline until the airfoil decelerates at the maximum pitch angle, $t^* = 4$. The slope is smoother thereafter while the airfoil stays steady, $t^* = 6$. A final steep decrease occurs before shedding terminates, which is more pronounced for the *constant* A_0 curve. As a result, the total vortex circulation predicted by both models evolve similarly, gently reducing the ascending slope, and are in good agreement with data extracted from CFD up to the instant of vortex shedding, which shows the only noticeably discrepancy between curves, due to the lower values in the discrete-vortex strength curve for the *constant* A_0 model. From the shedding point forward, a decay in vortex circulation is expected for CFD results, whereas the value provided by LDVM remains constant, as dissipation effects on the free vortex are not accounted for in the low-order model.

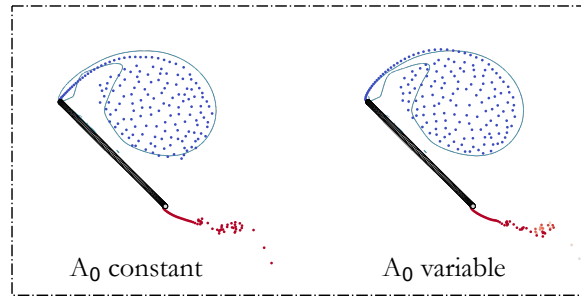


Fig. 5 Case I: Leading-edge vortex growth stage, $t^* = 5$. Rolling-up of discrete vortices from LDVM compared against the vortical structure defined in CFD by a $Q=1$ contour.

To help assess the accuracy of each LDVM version in reproducing the rolled-up configuration during the vortex growth phase, an overlapped image is shown in Fig. 5. The selected time instant, $t^* = 5$, corresponds to the midpoint of the motion. By that time, the leading-edge vortex already exhibits a distinctive fully-developed shape and still remains attached to the airfoil. An iso-contour of Q -criterion, $Q = 1$, was used to define the vortex in CFD, over which the vorticity contour from LDVM is superimposed for comparison. If the strength of vortex particles were calculated accurately, their mutual interaction should allow the roll-up into a vortex of the correct size and location. The good match observed in Fig. 5 confirms the success of LDVM in capturing this flow feature, independent of the vorticity shedding criteria implemented for cases where the airfoil thickness is small, the Reynolds number is low, and the unsteady motion such the formation of strong LEVs is evinced. The reason for the success of both LDVM implementations in these scenarios is the relatively low critical value of LESP, which makes the difference between the methods less significant.

B. Thick airfoil at low Reynolds number: NACA0012, $Re = 10000$

One of the strengths of the new formulation comes through the introduction of the thickness effect in the calculation of the rate of vorticity released at the leading edge. This second scenario, of thicker airfoils at low Reynolds numbers, can be found in lines of research focused on the application of flapping foils for bio-mimetic energy harvesting or underwater propulsion [59, 60].

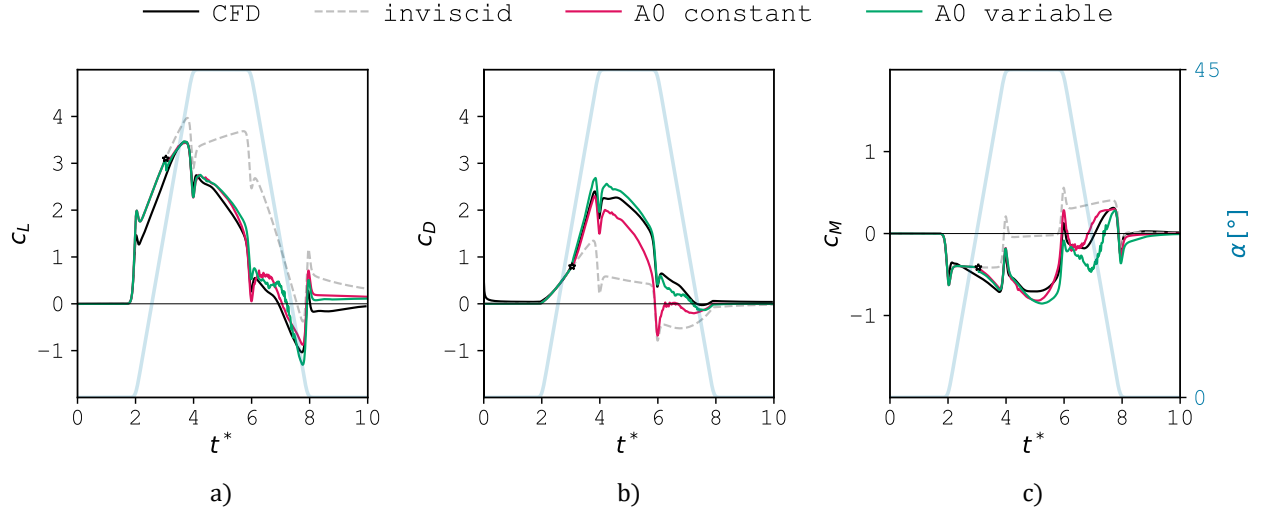


Fig. 6 Case II: aerodynamic coefficients. a) lift, b) drag, c) quarter-chord moment. The instant of LEV formation is indicated with a black marker. Also plotted for reference is the pitch angle profile.

Aerodynamic coefficients time histories are presented in Fig. 6. Lift comparison with results obtained from CFD reveals minor variations between models. With respect to the drag, both methods closely track the CFD curve during the upstroke. The approach of *constant* A_0 fails to keep a good agreement during the hold and downstroke parts of the motion. The *variable* A_0 formulation matches reasonably well. This improvement in drag force estimation with the new model could be of relevance to propulsive-efficiency oriented lines of research. Lastly, as it happened in the previous studied case, the moment coefficient during the downstroke exhibits a larger negative value than the CFD results, which could be related to the termination criterion adopted for the *variable* A_0 approach.

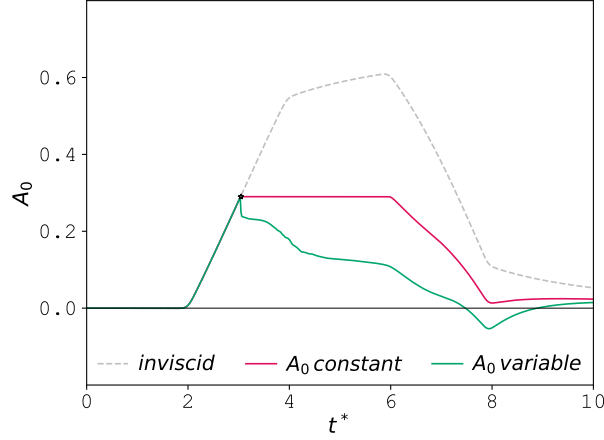


Fig. 7 Case II: Leading-edge suction parameter time history.

As in the former case, the evolution of the leading-edge suction parameter begins to diverge between approaches upon LEV formation, see Fig. 7. The decline for the *variable* A_0 curve during the ramp part of the motion is more abrupt for thicker airfoils. The negative slope then stabilizes during the hold segment, and increases again in magnitude during the return. After briefly reaching a negative value, it finally recovers the tendency displayed by the *constant* A_0 curve at the end of the motion.

Similar post-separation LESP trends have been reported in the computational studies by Narsipur et al. [40] for a NACA0012 airfoil pitching at its quarter chord with $K_\alpha = 0.4$ and $Re = 30000$, which are conditions similar to the ones in this subsection.

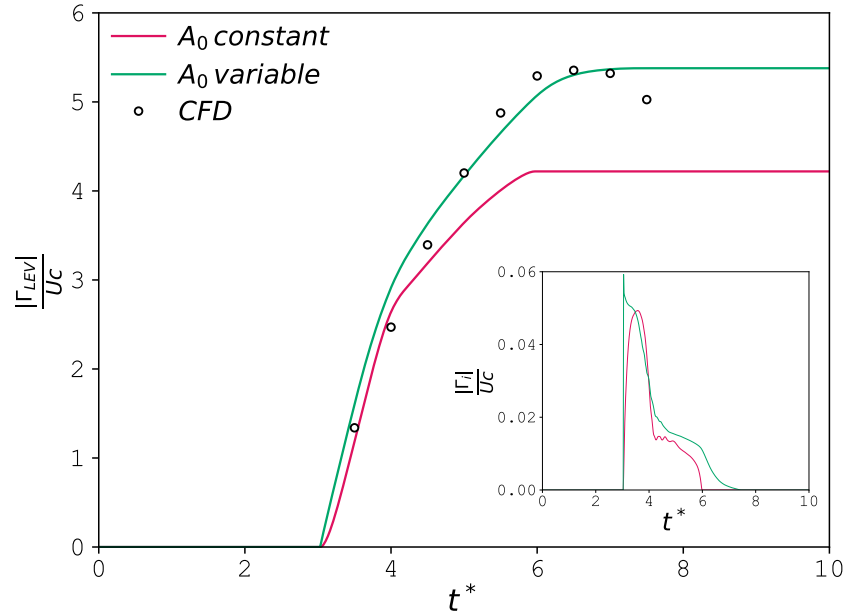


Fig. 8 Case II: Leading-edge vortex total circulation. Inset: nascent leading-edge discrete vortex strength.

Analyzing the strength of the leading-edge vortex through its total circulation and individual particle strengths, a remarkable dissimilarity between curves is evident after the airfoil reaches its maximum pitch angle, $t^* = 4$. The inset in Fig. 8 displays how the estimated strength of discrete vortices by the *constant* A_0 criterion drastically diminishes from this point. This results in the LEV total circulation curve drifting away considerably from the CFD value. On the

contrary, vortex circulation predicted by the *variable* A_0 method matches more closely the computational values at this stage of LEV growth. This demonstrates an improvement in the calculation of the strength assigned to each vortex particle when the velocity at the leading-edge shear layer is considered in the model.

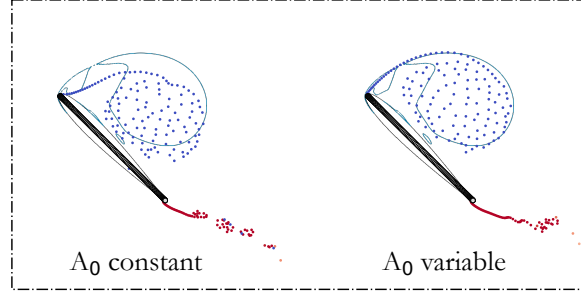


Fig. 9 Case II: Schematic of LEV roll-up during vortex growth stage, $t^* = 5$. A $Q=1$ contour is used to delimit the vortex in CFD, over which discrete vortices from LDVM are overlapped for comparison.

To reinforce these findings, vorticity contours are examined in Fig. 9 at the same time instant as shown for the case of thin airfoils, which is around 2 convective times after the onset of LEV formation. We observe that the slope of the separated shear layer and the location and shape of the LEV are better captured by the *variable* A_0 approach. This further illustrates that the vorticity shedding rate is more accurately calculated by the new method presented in this paper.

C. Thick airfoil at high Reynolds number and low reduced frequency: NACA0015, $Re = 550000$, $k = 0.15$ and 0.075

The last scenario considered in this paper aims to explore the capability of the new model in predicting post-stall behavior in situations where the hypotheses on which it is built are broken. Thick airfoils at high Reynolds numbers and low rates of motion are prone to trailing-edge flow separation prior to dynamic-stall vortex formation, and/or turbulence transition, phenomena not modeled in LDVM. These situations are often encountered in helicopter rotor blades and wind turbines. Additionally, the airfoil thickness is usually larger than in the cases previously studied, further questioning the effectiveness of the original version of LDVM.

For this purpose a NACA0015 airfoil undergoes harmonic pitching oscillations about its quarter chord according to the expression

$$\alpha(t) = 20^\circ \pm 8^\circ \sin(\omega t) \quad (14)$$

with reduced frequencies of $k = 0.15$ and $k = 0.075$.

This set of cases has been taken from experiments conducted by the UNFoLD lab at EPFL to study the onset of dynamic stall [51]. The data generated from these efforts serves to validate the low-order model in this scenario. No CFD results are included in this subsection, because the in-house version used for the rest of the paper was developed for low Reynolds number problems.

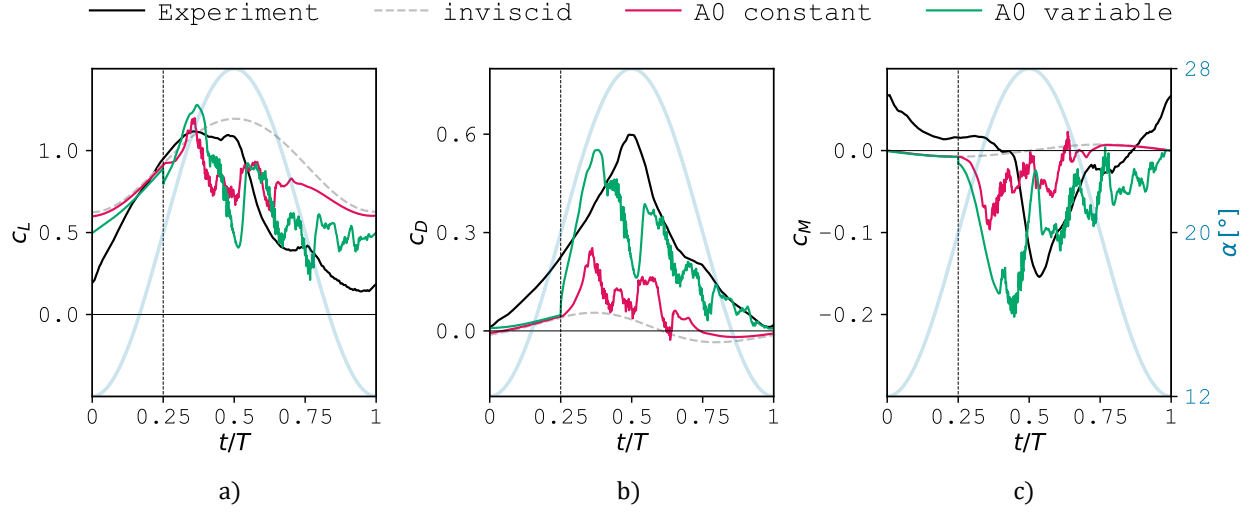


Fig. 10 Case III ($k = 0.15$): aerodynamic coefficients. a) lift, b) drag (pressure drag only), c) quarter-chord moment. A vertical dashed line represents the timing of static stall. The sinusoidal profile for the pitch angle is included as a reference.

Three important considerations should be kept in mind when comparing the numerical and experimental results. First, the experimental data provided is the average over 78 cycles, whereas numerical results are given for just one cycle of motion. Second, experiments were conducted in a wind tunnel with open test section, resulting in a lift slope lower than the theoretical value, and a factor of 0.435 is needed for comparison. Third, initiation and termination of leading-edge vortex shedding in LDVM are set to match experimental flow indicators. When the static-stall angle is exceeded shedding starts. When the experimental suction parameter becomes positive, after having dropped to negative values during vortex shedding, release of vorticity at the leading edge terminates.

Figure 10 shows the time history of aerodynamic coefficients for the case with the higher reduced frequency. Two clear peaks can be seen in the experimental lift curve at high angles of attack. The first one is the lift overshoot, hallmark of dynamic stall, experienced when the shear layer rolls-up into the dynamic-stall vortex. The second one is linked to its presence on the airfoil [51]. Both models seem to be capable of predicting the timing of the first peak with great accuracy, although not the second one. After the dynamic-stall lift overshoot, both models fail to reproduce the experimental tendency, and they present fluctuations that are dampened in time, with the mean value decreasing to recover its initial value. Regarding drag and moment coefficients, the *constant* A_0 approach gives very poor results. Because the A_0 value is larger than it should, so is the suction force. As a consequence the drag component of the normal force is reduced, to the point of becoming slightly negative by the end of the motion. The *variable* A_0 method predicts more successfully the magnitude and trend observed experimentally, although retaining the aforementioned damped fluctuations.

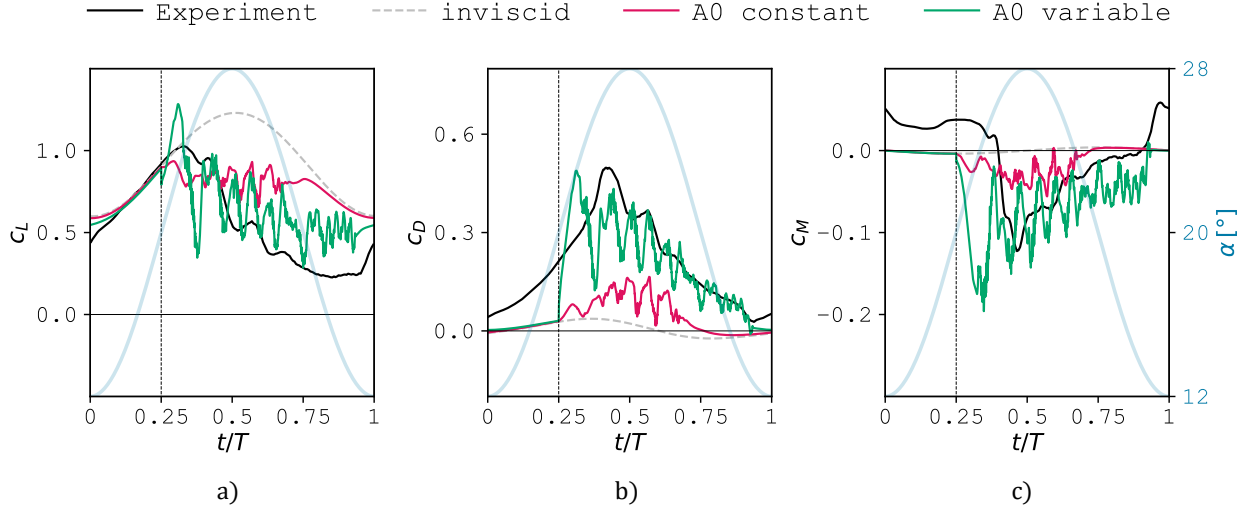


Fig. 11 Case III ($k = 0.075$): aerodynamic coefficients. a) lift, b) drag (pressure drag only), c) quarter-chord moment. The timing of static stall is indicated by a vertical dashed line. The pitch angle profile is also included for reference.

Figure 11 shows the time history of aerodynamic coefficients for the case with the lower reduced frequency. Reducing the pitch frequency does not seem to affect the general post-stall trends of aerodynamic coefficients significantly. Nonetheless, two facts are to be noted in Fig. 11: the timing of lift boost due to the dynamic-stall vortex is slightly less accurately predicted than in the case with higher reduced frequency, the discrepancy being more noticeable with the *constant* A_0 strategy. The second point to be stressed is that the frequency of loads fluctuations has been doubled by halving the reduced frequency. However, this is consistent with the fact that the airfoil is now exposed to dynamic stall for double the time.

It is also worth emphasizing the consequence of a wrong estimation in the circulation rate at the leading edge. An excessive prediction of suction force results in a lift increment and a drag reduction, and this is reflected in Fig. 11 when comparing the performance of both LDVM models against experimental data.

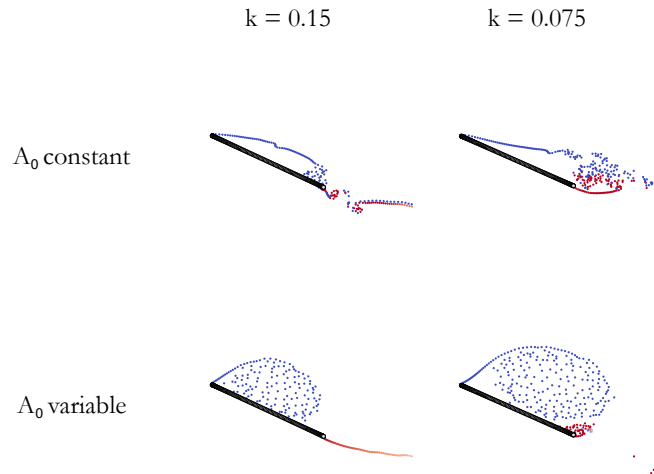


Fig. 12 Case III: Vorticity contours from LDVM at time instant of maximum lift coefficient.

The non-linearities observed in the aerodynamic loads are caused by vortices being shed into the wake. To visually support this observation, vorticity contours are shown in Fig. 12 for the time instant at which lift coefficient is maximum in experimental data. This corresponds to $t/T = 0.36$ for $k = 0.15$, and $t/T = 0.33$ for $k = 0.075$. The intention of this figure is twofold: to illustrate the success of the new model in reproducing dynamic-stall vortex rolling-up, against the difficulty faced by the original model; and to serve as a visual complement to the aforementioned loads fluctuations. An incipient trailing-edge vortex (TEV) for $k = 0.075$, not yet visible for $k = 0.15$, is indicative of the faster rate at which the LEV-TEV arrangement is occurring at lower reduced frequencies. This pair is ultimately responsible for the oscillations observed in loading.

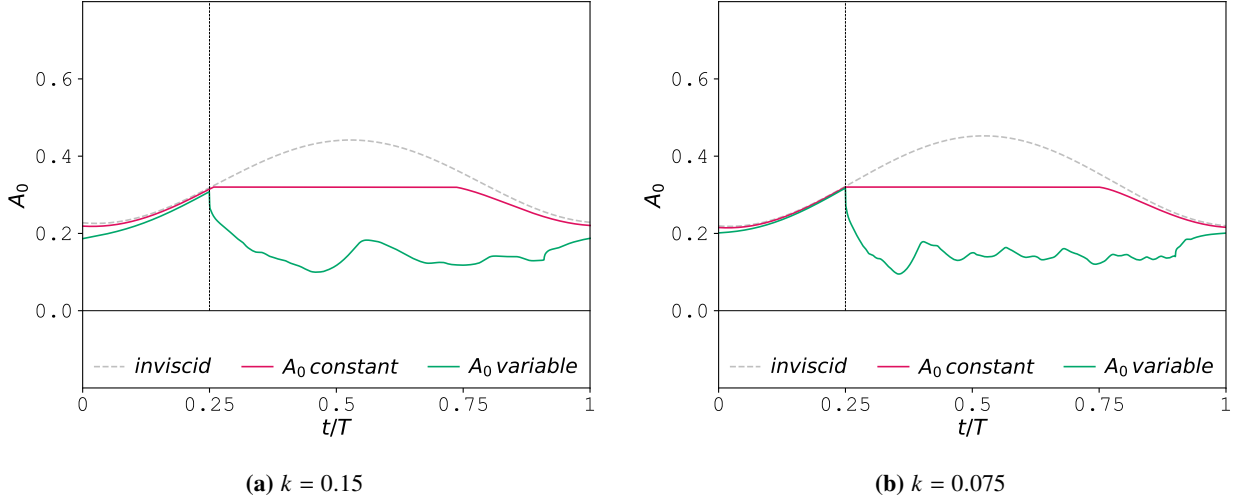


Fig. 13 Case III: Leading-edge suction parameter time history.

An analysis of the leading-edge suction parameter from the two LDVM methods is carried out in Fig. 13. Similarly to what happened with the ramp cases, suction parameter decays after the vortex starts forming. However, as opposed to the ramp scenarios in Sec. V.A and Sec. V.B, it does not fall to values close to zero at any time of the motion. After the initial drop, with a more gentle slope for the fastest case, the LESP starts oscillating around a constant mean value which is kept for the rest of the motion. Although the timing of fluctuations does vary with the reduced frequency, for the reason commented before, the magnitude of this constant value barely changes.

VI. Conclusions

The broad aim of this paper has been to improve the performance of augmented low-order models, like LDVM, in simulating LEV-dominated unsteady flow problems. We presented a new formulation to estimate the rate of circulation fed into the leading-edge vortex more accurately. The strength of discrete vortices shed from the leading edge, when the LESP critical value is exceeded, is here calculated based on the velocity at the edge of the separated shear layer. CFD and experimental results are included to test the accuracy of this new LDVM version and the original one to model leading-edge vortices in various scenarios.

For thin airfoils operating at low Reynolds number and fast rates of motion, differences between models are minor. But when the thickness of the airfoil is increased, these discrepancies become more noticeable. Even when the original LDVM captures fairly nicely the time history of load coefficients, it fails to reproduce the correct evolution of LEVs. This is manifested in an underestimation of the total vortex circulation, and an incorrect roll-up of the vortical structure. By contrast, the new approach to calculate the rate of circulation at the leading edge is able to capture the observed decay in LESP during vortex shedding. Therefore it provides a more precise estimation of the leading-edge vortex circulation, and hence a correct vortex growth behavior.

On the other hand, for the higher Reynolds number and lower rate of motion cases, because trailing-edge separation is not modeled in these LDVM versions, results provided are in less agreement with experimental data. Still, they are able to capture the stall delay. Furthermore, the new model reproduces the roll-up of the dynamic-stall vortex, which the original version does not.

Overall, our new model has the potential to enhance the performance in unsteady flow problems which involve the formation of strong LEVs, increasing the range of applicability to thicker airfoils and higher Reynolds numbers.

Acknowledgments

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